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Maximum likelihood estimators and worst case optimal algorithms for system identification

R. TEMPO *

CENS-CNR, Politecnico di Torino, Corso Duca degli Abruzzi 24, 10129 Torino, Italy

G.W. WASILKOWSKI **

Department of Computer Science, University of Kentucky, Lexington, KY 40506, U.S.A.

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Abstract: For system identification problems with stochastic noise, maximum likelihood estimators are frequently used. If noise is deterministic, worst case optimal algorithms should be considered. In this paper we study the following problem: Under what circumstances are maximum likelihood estimators optimal in the worst case?

Keywords: System identification, Noisy information, Maximum likelihood estimators, Optimal algorithms, Worst case.

1. Introduction

There is a vast literature studying system identification problems from a stochastic point of view. It is assumed that the observations are contaminated by an additive noise which is *random*. Depending on a problem, different estimators are proposed. For a number of reasons, including asymptotic optimality (see e.g. [1]), *maximum likelihood estimators* (*MLE*) are commonly used. The stochastic approach requires at least partial knowledge of noise distribution. Although this is not a subject of this note, we stress that in practice there is an obvious difficulty in getting such knowledge.

There is an alternative approach, called *worst case*, which we discuss briefly now (see also [2–5]). In the worst case, the noise is considered to be deterministic, and instead of its distribution an upper bound on the noise is supposed to be known. In this approach, one tries to find optimal algorithms which are defined as to minimize the maximal (with respect to the noise) distance between actual and computed parameters for the worst data. Strongly optimal algorithms are optimal algorithms for all possible data.

Stochastic and worst case approaches are different: Worst case approach is based on the assumption that the noise is very malicious, and whatever estimator is used, the worst possible noise will occur. Hence, the worst case is more pessimistic than the stochastic one. On the other hand, the worst case guards against the most difficult case which, even if not very likely, can happen. This means that a stochastic estimator can commit an error which is much larger than the one of a worst case optimal algorithm, unless the estimator equals the optimal algorithm.

Thus, it is important to know:

• *Under what assumptions does there exist an algorithm which is optimal from both stochastic and worst case point of view?*

In this paper we attempt to answer this question by studying linear problems and analyzing worst case optimality of Maximum Likelihood Estimators for a number of distributions of noise such as uniform, normal and Laplace. We also investigate some robustness properties of Maximum Likelihood Estimators introducing the class of robust-interpolatory algorithms. The formal definition of these algorithms is presented in Section 2. Here we only mention that any robust-interpolatory algorithm is 'almost' optimal (i.e. it is optimal modulo a multiplicative constant), but this holds independently of the upper bound on the noise, whereas optimal algorithms depend on the specific bound and may not be optimal for other bounds on the noise.

In particular, we show:

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- For uniform distribution: in general, MLE's are neither optimal nor robust-interpolatory, though there exist two MLE's which are optimal or robust-interpolatory, if the noise is bounded in l_∞ norm.
- For normal distribution: the unique MLE is robust-interpolatory and optimal, if the noise is bounded in l_2 norm.
- For Laplace distribution: the unique MLE is robust-interpolatory, if the noise is bounded in l_1 norm, but is not optimal.

2. Problem formulation

In this section we present formal definitions and notations concerning the worst case. For a more detailed discussion of the general theory see [2–5]; for specific applications to system parameter estimation, time series prediction and filtering see [5–7].

Let X and Z be finite dimensional linear spaces over the real field $\Re$. We assume that Z is equipped with a norm $\|\cdot\|_Z$. Let S, called a *solution operator*, be a linear mapping

$$S: X \to Z.$$

Our aim is to estimate Sx for every $x \in X$ knowing approximate information about x. Let N, called *information operator*, be a linear mapping

$$N: X \to Y,$$

where Y is a normed linear space over $\Re$ with a norm $\|\cdot\|_Y$. In this paper we assume that the information operator N is a complete information, i.e. it is a one-to-one mapping. In general, in the presence of noise, the exact information on x is not available and only perturbed information y is given

$$y = Nx + \rho,$$

where ρ is the actual noise contaminating Nx.

An approximation z to Sx is constructed by an *algorithm* (or *estimator*) A using the available data y,

$$z := A(y),$$

where A is an arbitrary mapping

$$A: Y \to Z.$$

In the *stochastic* approach, the actual noise ρ is assumed to be a realization of a random variable with a given density ρ. That is, $Y = \Re^m$ and

$$\Pr(\rho \in B) = \int_B p(\rho)\, \mathrm{d}\rho,$$

for every Borel subset B of Y. Here d stands for Lebesgue measure. Maximum Likelihood Estimators (MLE) are among the most commonly used. They are defined as follows. Let $p(y \mid x) = p(y - Nx)$. An estimator A is MLE iff for every $y \in Y$,

$$A(y) = Sx^{\mathrm{MLE}},$$

where $x^{\mathrm{MLE}}(y)$ is so that

$$p(y \mid x^{\mathrm{MLE}}) = \max_{x \in X} p(y \mid x). \tag{1}$$

Any MLE is denoted by A^{MLE}. As mentioned in the Introduction, we would like to know when MLE are strong optimal (or 'almost' strongly optimal) in the worst case setting, which is now discussed.

In the *worst case* setting, the actual noise ρ is deterministic, and we assume that it never exceeds a given upper bound ε. That is,

$$\|\rho\|_Y \le \varepsilon.$$

Thus, given information y, we know that the element x satisfies $y = Nx + \rho$ with $\|\rho\|_Y \le \varepsilon$. That is,

$$x \in T(y, \varepsilon) := \{x \in X: \|y - Nx\|_Y \le \varepsilon\}.$$

Hence, given y, the *worst case error* of an algorithm A is equal to

$$e(A, y) = \sup_{x \in T(y,\varepsilon)} \|Sx - A(y)\|_Z.$$

By a *strongly optimal algorithm* we mean any A° whose error is minimal for all data y with nonempty $T(y, \varepsilon)$, i.e.

$$e(A^\circ, y) = \inf_A e(A, y), \quad \forall y.$$

Note that the minimal error, given y, is attained by an algorithm whose value at y coincides with the Chebyshev center of the set $S(T(y, \varepsilon))$. Recall that $c = c(y)$ is a center of $S(T(y, \varepsilon))$ iff

$$\sup_{z \in S(T(y,\varepsilon))} \|z - c\|_Z = \inf_{\alpha \in Z} \sup_{z \in S(T(y,\varepsilon))} \|z - \alpha\|_Z.$$

For simplicity we assume that such a center exists for every y for which $T(y, \varepsilon)$ is nonempty. Then

the *central* algorithm A^c is defined by

$$A^c(y) = c(y). \tag{2}$$

Obviously, an algorithm is central iff it is strongly optimal. Due to optimality properties of central algorithms, we are interested in efficient ways of constructing them (see [5,8]). Unfortunately, central algorithms may be difficult to implement in some cases. For this reason different classes of algorithms, including interpolatory and projection algorithms (see [2,8]) have been studied in the literature.

An algorithm A^I is *interpolatory* iff for every y with nonempty $T(y, \varepsilon)$

$$A^I(y) = Sx^* \quad \text{for some } x^* \in T(y, \varepsilon). \tag{3}$$

It is known, see e.g. [2,3], that interpolatory algorithms are strongly optimal to within a factor of 2. That is,

$$e(A^I, y) \le 2e(A^c, y) = 2 \inf_A e(A, y), \quad \forall y.$$

In general, interpolatory algorithms are not strongly optimal, but they are 'almost' strongly optimal. Furthermore, for some problems there exists a relatively easy to compute element $x^* \in T(y, \varepsilon)$, and so is the interpolatory algorithm $A^I(y) = Sx^*$.

Note that in the definitions of central and/or interpolatory algorithms the bound ε on the noise is fixed. In particular, this means that optimality of algorithms depends on ε. Thus, it could happen that an algorithm is interpolatory (or even central) for a particular value of ε, but is not interpolatory for other values. This is undesirable in a number of situations including the following ones.

- Sometimes the sharp bound on the noise is difficult (or even impossible) to get, and ε is not tight.
- Even if ε is tight, it is tight in the worst case sense, and it could happen that the actual noise ρ has norm much smaller than ε.

Both cases describe situations when the actual noise ρ is smaller than ε and it is possible to approximate Sx very well, though central and/or interpolatory algorithms do not 'feel' the smallness of ρ and propose an approximation which is optimal (or almost optimal) in the worst case, i.e. when $\rho \approx \varepsilon$. Thus, in general, central and/or interpolatory algorithms are not *robust* with respect to the bound ε.

Therefore, we are also interested in robust-interpolatory algorithms. An algorithm A^R is *robust-interpolatory* iff it is interpolatory for all positive $\varepsilon_0 \le \varepsilon$. That is,

$$A^R(y) = Sx^R \quad \text{for some } x^R \in T(y, \varepsilon_0),$$
$$\|\rho\| \le \varepsilon_0.$$

It is easy to see that A^R is robust-interpolatory iff

$$\|Nx^R(y) - (Nx + \rho)\|_Y \le \|\rho\|_Y,$$
$$\forall \|\rho\|_Y \le \varepsilon, \quad \forall x \in X. \tag{4}$$

Note (by letting $\varepsilon \to 0$) that to be robust-interpolatory an algorithm should be *correct*, i.e.

$$A(Nx) = Sx;$$

see [5] for some optimality properties of correct algorithms.

We end this section showing that robust-interpolatory algorithms coincide with projection algorithms. Recall that A^P is a *projection* algorithm iff $A^P(y) = Sx^P$ for x^P being a solution of

$$\|y - Nx^P\|_Y = \min_{\tilde{x}} \|y - N\tilde{x}\|_Y. \tag{5}$$

Lemma 1. *Robust-interpolatory and projection algorithms coincide. That is, every projection algorithm is robust-interpolatory, and for every robust-interpolatory A^R,*

$$A^R(y) = A^P(y) \tag{6}$$

whenever $T(y, \varepsilon) \ne \emptyset$.

Proof. It is an easy exercise to prove that A^P is robust-interpolatory. Consider therefore an arbitrary robust-interpolatory algorithm A^R. For given y, let x^P be as in (5). Take now $\tilde{\rho} = y - Nx^P$. Obviously, $\|\tilde{\rho}\|_Y \le \varepsilon$ and therefore, (4) yields that

$$\|Nx^R - y\|_Y \le \|\tilde{\rho}\|_Y.$$

Since $\|\tilde{\rho}\|_Y = \min_{\tilde{x}} \|y - N\tilde{x}\|_Y$, this proves that A^R is equal to a projection algorithm. □

Finally, notice that if $\|\cdot\|_Y$ is a Hilbert norm, the projection algorithm is central while for non-Hilbert norms $\|\cdot\|_Y$, such as l_1 or l_∞, it is not central (see [8]).

3. Maximum likelihood estimators and worst case optimal algorithms

In this section we attempt to answer the question when MLE are robust-interpolatory or central. We begin with the general case of arbitrary distribution of noise and arbitrary norms in Y.

Lemma 2. *For arbitrary distribution of noise with unique* MLE,

$$A^{\mathrm{MLE}}(Nx+\rho) = Sx + A^{\mathrm{MLE}}(\rho).$$

Proof. Let $Sr^{\mathrm{MLE}} = Sx^{\mathrm{MLE}}(\rho) = A^{\mathrm{MLE}}(\rho)$, i.e.

$$p(\rho - Nr^{\mathrm{MLE}}) = \max_{\tilde{x}} p(\rho - N\tilde{x}).$$

Using the definition (1) of A^{MLE} we get

$$\begin{aligned} &p(Nx - Nx^{\mathrm{MLE}} + \rho) \\ &\quad = p(-N(x^{\mathrm{MLE}} - x) + \rho) \\ &\quad = \max_{\tilde{x}} p(-N(\tilde{x} - x) + \rho) \\ &\quad = \max_{r} p(\rho - Nr) = p(\rho - Nr^{\mathrm{MLE}}). \end{aligned}$$

This implies

$$r^{\mathrm{MLE}} = x^{\mathrm{MLE}} - x,$$

and completes the proof. □

Remark 1. Lemma 2 still holds when MLE's in A^{MLE} are not unique. Then $A^{\mathrm{MLE}}(y)$ denotes the set of values of all MLE's at y and $+$ denotes the algebraic sum.

From Lemma 2 we immediately get:

Proposition 1. *For arbitrary distribution of noise, A^{MLE} is robust-interpolatory iff there exists positive ε such that*

$$\| Nx^{\mathrm{MLE}}(\rho) - \rho \|_Y \le \| \rho \|_Y, \quad \forall \| \rho \|_Y \le \varepsilon.$$

Unfortunately, the condition presented in Proposition 1 cannot be easily checked for a given distribution. However, a general link between probability density functions and norms in Y is simply shown in:

Proposition 2. *Let the distribution of noise satisfy*

$$p(\rho) = h(\| \rho \|_Y),$$

for a decreasing function $h: \Re^+ \to \Re^+$. Then A^{MLE} is a projection algorithm and, consequently, robust-interpolatory.

Now we comment on specific distributions of noise.

3.1. Uniform distribution of noise

Let $Y = \Re^m$. For positive v_i, $i = 1, \ldots, m$, let

$$K = [-v_1, v_1] \times \cdots \times [-v_m, v_m].$$

We assume that

$$p(\rho) = \begin{cases} (2^m v_1 \ldots v_m)^{-1} & \text{if } \rho \in K, \\ 0 & \text{otherwise.} \end{cases}$$

Then (1) yields that Sx is a value of MLE iff

$$|(Nx)_i - y_i| \le v_i, \quad i = 1, \ldots, m. \tag{7}$$

Condition 1. Let Y be equipped with l_∞^w norm with weights $w_i = \varepsilon / v_i$ and let $Z = \Re^r$ be equipped with l_∞ norm, i.e.

$$y = [y_1, \ldots, y_m]^{\mathrm{T}} \quad \text{and} \quad \| y \|_Y = \max_i |y_i| w_i,$$

$$z = [z_1, \ldots, z_r]^{\mathrm{T}} \quad \text{and} \quad \| z \|_Z = \max_i |z_i|.$$

Then (7) is equivalent to

$$\| Nx - y \|_Y \le \varepsilon.$$

Hence, in this setting, any MLE is interpolatory and, consequently, 'almost' strongly optimal but not necessarily central nor robust-interpolatory. Note that there are infinitely many maximum likelihood estimators for uniform distribution, and different estimators were proposed in the literature. For instance, among most commonly proposed is the following one (see [9]):

$$A^{\mathrm{MLE}}(y) = Sx^{\mathrm{MLE}}, \tag{8}$$

with x^{MLE} being the solution of the problem

$$\| Nx^{\mathrm{MLE}} - y \|_Y = \min_{\tilde{x}} \| N\tilde{x} - y \|_Y.$$

Note that this problem is equivalent to

$$\begin{aligned} \min\{\delta:\ & w_i(Nx)_i - \delta \le w_i y_i, \\ & w_i(Nx)_i + \delta \ge w_i y_i, \\ & \delta \ge 0,\ i = 1, \ldots, m\}, \end{aligned}$$

which is a linear programming problem with $2m+1$ constraints and a number of parameters equal to $\dim X+1 \le m+1$. Thus, to evaluate this MLE one needs to find $\tilde{x}$ (by solving the above linear programming problem) and apply S to $\tilde{x}$. As this algorithm is a projection algorithm, it is robust-interpolatory, but it is not central. Still in the same setting, we exhibit an MLE which is central. From [5] we have the following formula for the central algorithm A^c.

$$A^c(y) = [a_1, \ldots, a_r]^T, \tag{9}$$

where

$$a_i = \tfrac{1}{2}\sup\left\{(Sx)_i : |(Nx)_j - y_j| \le \frac{\varepsilon}{w_j},\ j=1,\ldots,m\right\} + \tfrac{1}{2}\inf\left\{(Sx)_i : |(Nx)_j - y_j| \le \frac{\varepsilon}{w_j},\ j=1,\ldots,m\right\}.$$

Thus, the central algorithm (9) can be computed by solving $2r$ linear programming problems, each of them with $2m$ constraints and the number of parameters equal to $\dim X \le m$.

For further optimality properties of central algorithms see [10], and for results showing numerical comparisons between central algorithms and classical statistical estimators see [6]. Furthermore, some relations between estimators for worst case and MLE can be found in [11].

We summarize this in:

Proposition 3. *For uniformly distributed noise under Condition* 1:

- *Every MLE is almost strongly optimal, but in general not robust-interpolatory.*
- *MLE given by* (7) *is robust-interpolatory, but not central.*
- *MLE given by* (8) *with* $w_i = \varepsilon/v_i$ *is central.*

3.2. Normal distribution of noise

Let $Y = \Re^m$ be equipped with a Hilbert norm $\|\cdot\|_Y$ and $Z = \Re^r$ with an arbitrary norm $\|\cdot\|_Z$. We assume that

$$p(\rho) = \frac{1}{\sqrt{(2\pi)^m |\Sigma|}} \exp\left(-\tfrac{1}{2}\rho^T \Sigma^{-1} \rho\right), \quad \forall \rho \in \Re^m,$$

where Σ is a symmetric positive definite covariance matrix. In this case, the unique MLE is

$$A^{\mathrm{MLE}}(y) = Sx^{\mathrm{MLE}},$$

where x^{MLE} is given by the Gauss–Markov formula

$$x^{\mathrm{MLE}} = (N^T \Sigma^{-1} N)^{-1} N^T \Sigma^{-1} y.$$

We now discuss central algorithms. From [8] we know that the central algorithm is equal to the least squares estimator,

$$A^c(y) = Sx^*,$$

where x^* satisfies

$$\|Nx^* - y\|_Y = \min_{\tilde{x}} \|N\tilde{x} - y\|_Y.$$

Suppose now that

$$\|y\|_Y = \sqrt{y^T \Sigma^{-1} y}. \tag{10}$$

Then

$$x^* = (N^T \Sigma^{-1} N)^{-1} N^T \Sigma^{-1} y,$$

and the central algorithm coincides with the MLE. This algorithm is also interpolatory. Note that x^* is independent of ε. Hence, it is central for any ε and robust-interpolatory, which also follows from Proposition 2.

We summarize this in:

Proposition 4. *For normally distributed noise, MLE is central and robust-interpolatory, if* $\|\cdot\|_Y$ *is given by* (10).

3.3. Laplace distribution of noise

Let $Y = \Re^m$ be equipped with the l_1 norm, i.e.

$$\|y\|_Y = \sum_{i=1}^{m} |y_i|.$$

Let

$$p(\rho) = \left(\tfrac{1}{2}c\right)^m \exp\left(-c\sum_{i=1}^{m} |\rho_i|\right), \quad \forall \rho \in \Re^m,$$

where c is a positive number. Then the unique MLE is given by

$$A^{\mathrm{MLE}}(y) = Sx^{\mathrm{MLE}},$$

with $x^{\mathrm{MLE}}(y)$ being the solution of a minimum norm problem

$$\| Nx^{\mathrm{MLE}} - y \|_Y = \min_{\tilde{x}} \| N\tilde{x} - y \|_Y .$$

Note that computation of x^{MLE} corresponds to a linear programming, but the number of constraints is equal to $2^m + 1$. The MLE is robust-interpolatory, as it is a projection algorithm. However, it is not central for a number of problems (i.e. specific X, Z, S and N), including (see [8])

$$X = Z = \Re, \quad S = I, \quad Nx = [x, 2x]^{\mathrm{T}}.$$

We summarize this in:

Proposition 5. *For Laplace distribution of noise the MLE is robust-interpolatory, but it is not central.*

4. Conclusions

In this note we investigated some relations between optimal algorithms in a worst case sense and maximum likelihood estimators for some distributions of noise. In particular, for uniform and Laplace distributions, we have shown that maximum likelihood estimators are 'almost' optimal algorithms and that they enjoy some robustness properties with respect to the bound on the noise. On the contrary, the so-called central algorithms do not satisfy such robustness properties but they are optimal algorithms. However, for Gaussian distributions central algorithms coincide with maximum likelihood estimators and both requirements are satisfied. Finally, for general distributions we present conditions under which maximum likelihood estimators enjoy both optimality and robustness properties.

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